

TinyStories TiDAR Greedy Acceptance Comparison

All runs branch from the stable run at step 90,000:

- **Stable (blue)**: $\alpha=1$, $\beta=1$ – baseline AR+Diff training, continued to 121k
- **KL-only (red)**: $\alpha=0.2$, $\beta=0$, $\rho=0.5$, $\delta=0.3$ – forward KL + greedy, no diff loss
- **KL-keep (green)**: $\alpha=1$, $\beta=1$, $\rho=0.1$, $\delta=0.3$ – forward KL + greedy, keeps AR+Diff
- **Distill (purple)**: $\alpha=1$, $\beta=1$, $\eta=0.04$, $T=2$ – soft distillation AR->Diff
- **Greedy Eta (orange)**: $\alpha, \beta=0.1$, $\delta=3.0$, $\eta=1.0$, $T=0.7$ – aggressive agreement + distill
- **Stable bigger beta (teal)**: $\alpha=1$, $\beta=5$ – baseline with 5x diffusion weight
- **Top-K set (brown)**: $\gamma=0.01$, $\gamma_{\text{topk}}=8$ – top-k set distillation (later-stage)
- **Big Gamma+Delta (indigo)**: $\gamma=5$, $\gamma_{\text{topk}}=4$, $\delta=1$ – strong top-k + hard agreement

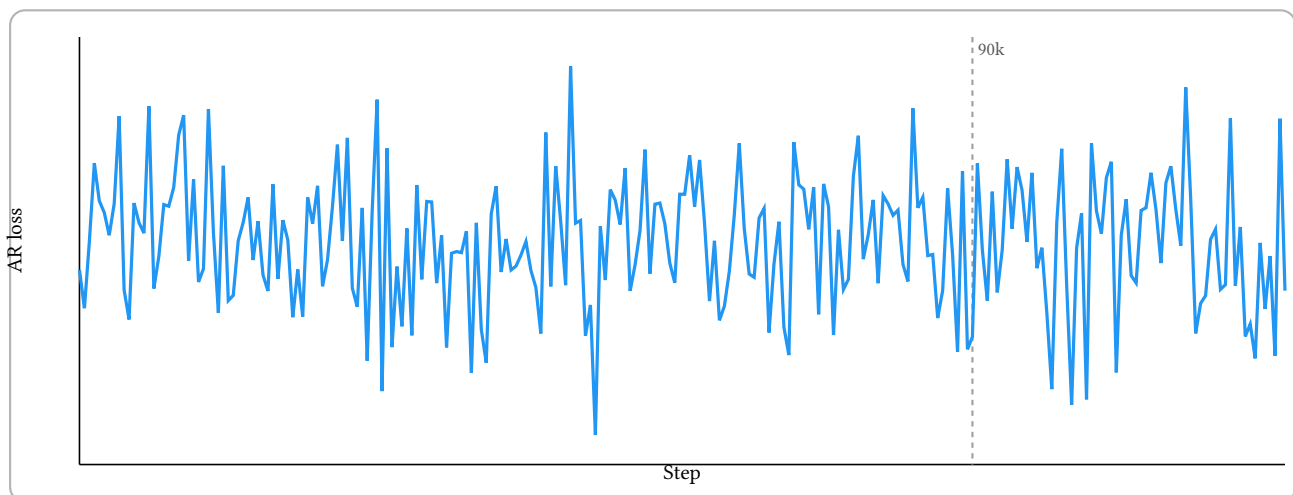
Legend

■ Stable (1, 1, 0, 0, 0, 0)	■ KL-keep (1, 1, 0.1, 0, 0.3, 0)	■ Greedy eta (0.1, 0.1, 0, 0, 3.0, eta=1.0 T=0.7)	■ Top-K set (gamma=0.01 K=8)
■ KL-only (0.2, 0, 0.5, 0, 0.3, 0)	■ Distill (1, 1, 0, 0, 0, eta=0.04 T=2)	■ Stable bigger beta (1, 5, 0, 0, 0, 0)	■ Big Gamma+Delta (gamma=5 K=4 delta=1)

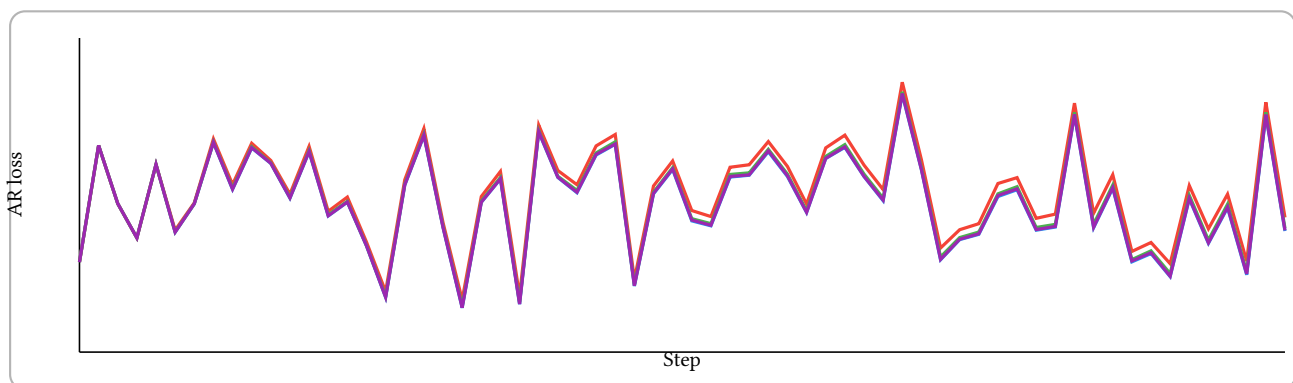
Total cost of the experiment so far: 28.40\$

AR loss (raw)

Stable 0-121k

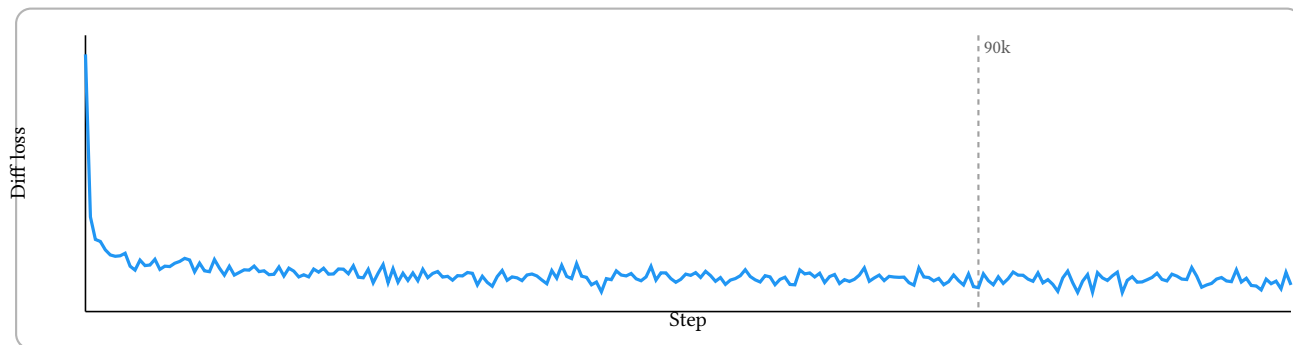


All runs 90k-121k

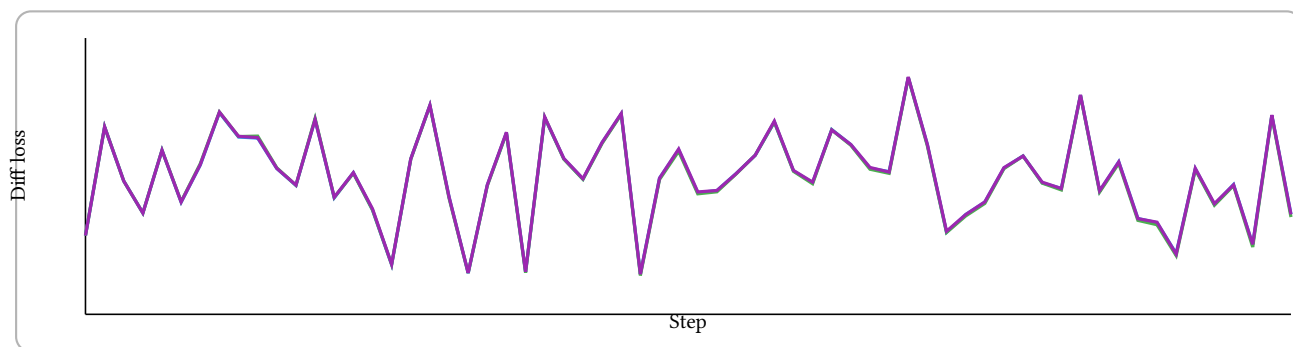


Diff loss (raw)

Stable 0-121k



All runs 90k-121k

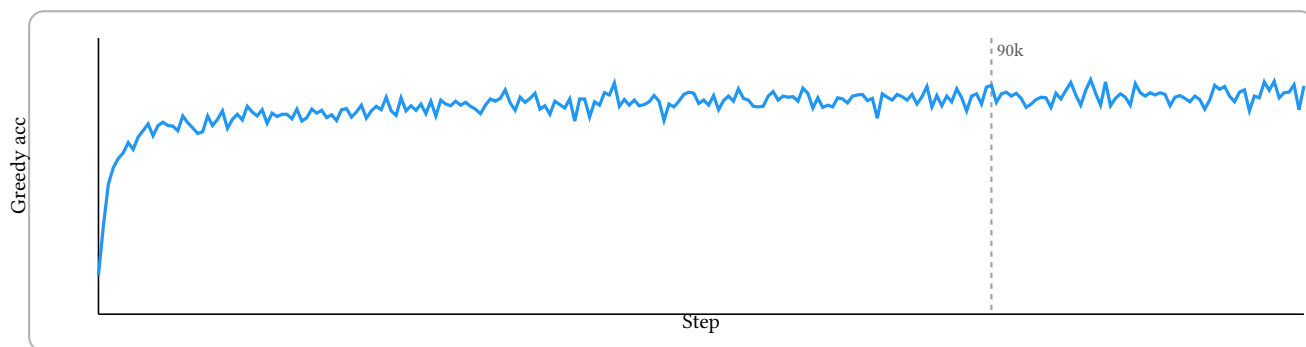


KL-only diff=0 after 90k, excluded from the zoomed panel.

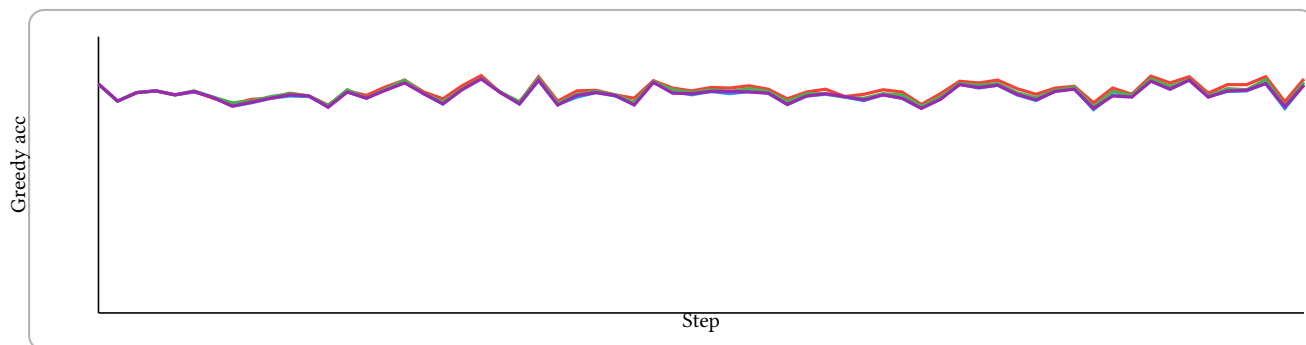
Greedy acceptance

Greedy acceptance = fraction of positions where $\text{argmax}(\text{AR}) == \text{argmax}(\text{Diff})$.

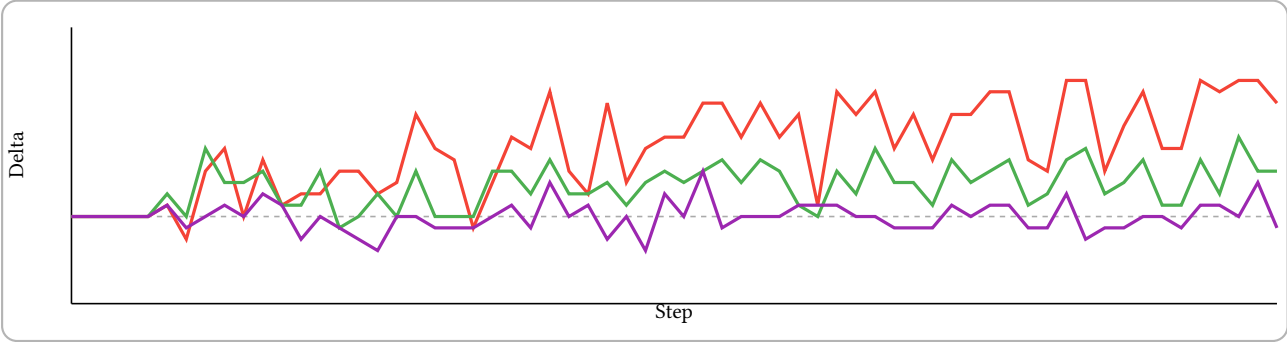
Stable 0-121k



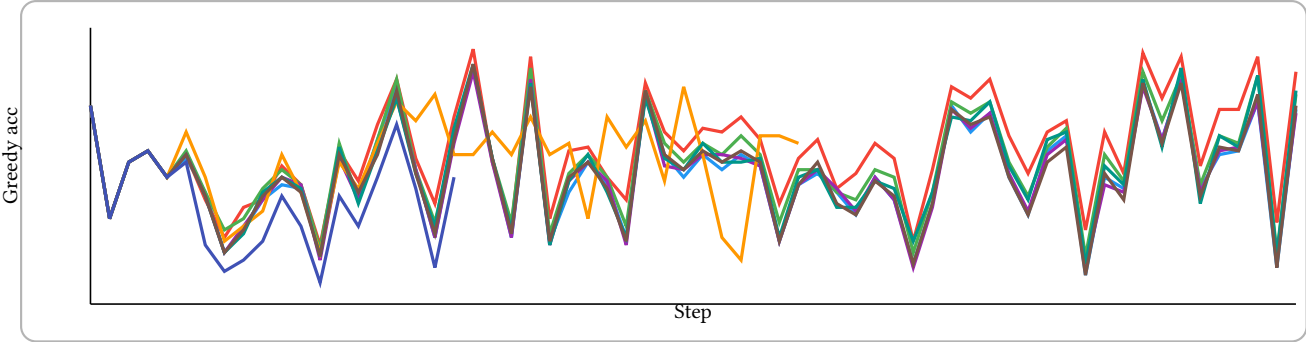
All runs 90k-121k



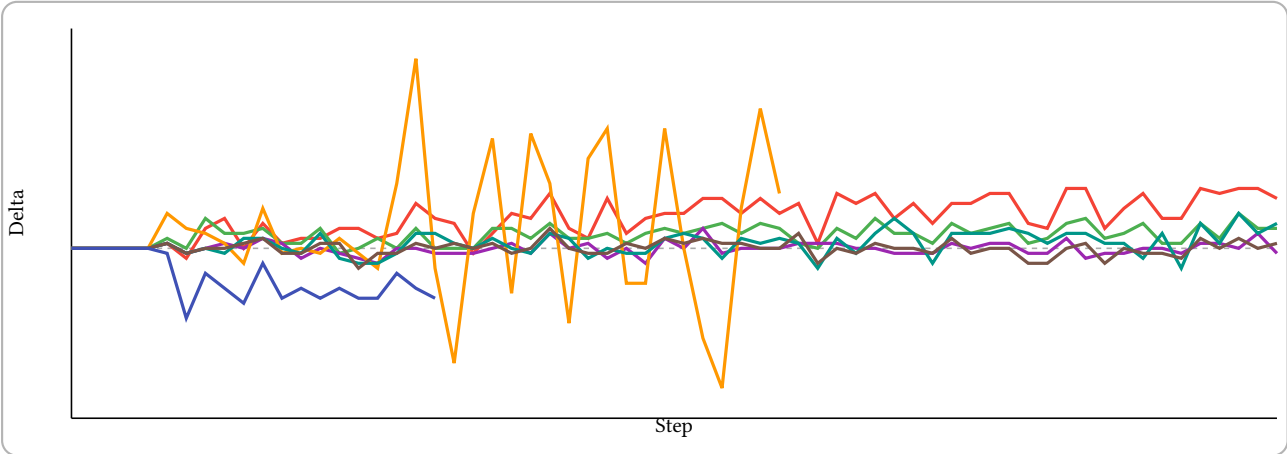
Delta vs stable (90k-121k)



All runs 90k-121k (incl. smallAR big greedy eta + stable bigger beta + top-k set + big gamma+delta)



Delta vs stable (90k-121k) (incl. smallAR big greedy eta + stable bigger beta + top-k set + big gamma+delta)



Inference Results

Greedy decoding (temperature=0) with draft_len=6.

Prompts (10):

- P0: “There was a small village”
- P1: “Once upon a time, a little girl”
- P2: “The boy was very happy because”
- P3: “In the forest, a tiny fox”
- P4: “The cat and the dog were”
- P5: “One day, the teacher said”
- P6: “After school, the kids went”
- P7: “The robot wanted to learn”
- P8: “On a sunny morning, Emma”
- P9: “The brave knight looked at”

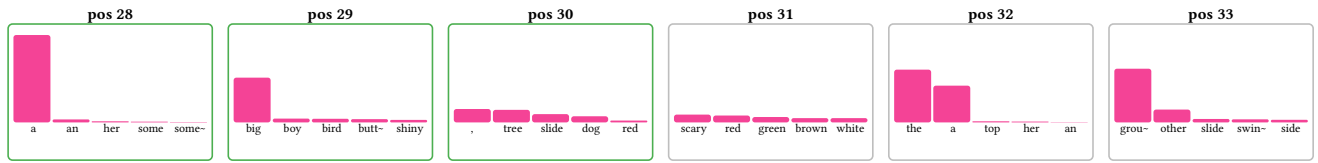
Accept/Iter summary (mean / best / worst across prompts)

Checkpoint	Steps=50	Steps=100	Steps=300
Stable @90k	2.41 / 3.50 / 1.63	2.26 / 2.75 / 1.71	2.10 / 2.45 / 1.88
Stable @121k	2.22 / 3.27 / 1.53	2.09 / 2.68 / 1.60	2.10 / 2.69 / 1.57
KL-only @121k	2.42 / 3.27 / 1.81	2.30 / 2.68 / 1.80	2.08 / 2.49 / 1.61
KL-keep @121k	2.40 / 3.27 / 1.75	2.29 / 2.83 / 1.80	2.18 / 2.90 / 1.75
Distill @121k	2.31 / 3.50 / 1.53	2.26 / 3.09 / 1.87	2.19 / 2.74 / 1.80
SmallAR big greedy eta @105k	2.15 / 2.88 / 1.75	2.15 / 2.75 / 1.90	2.00 / 2.43 / 1.74
Stable bigger beta @121.5k	2.37 / 3.27 / 1.63	2.33 / 3.09 / 1.62	1.96 / 2.27 / 1.74

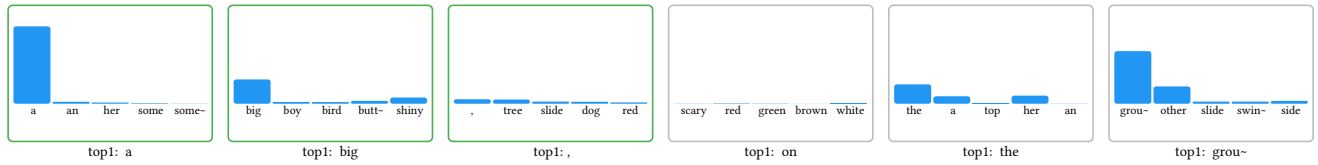
Inference Logits Histogram

Prompt: "Once upon" | Checkpoint: "step_0121566.npz" Target position: 30 | Block: 28-33 | Draft len: 6

AR verify logits (top-5, sorted by AR)



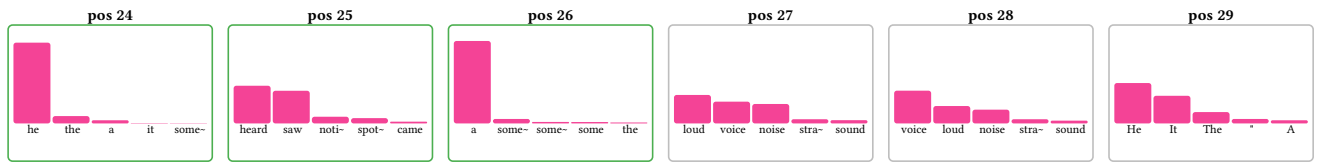
Diff draft logits (same tokens, AR order)



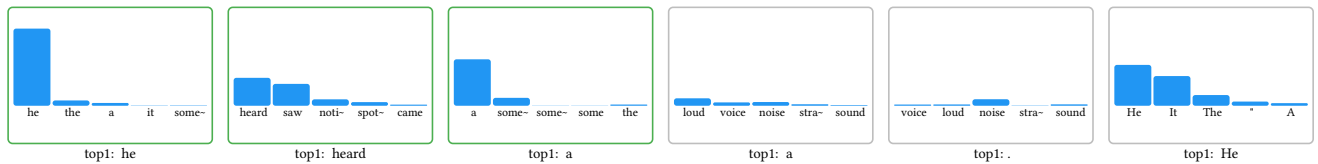
Sentence context One day, she saw a big, scary dog.

Prompt: "In the forest, a tiny fox" | Checkpoint: "step_0121566.npz" Target position: 27 | Block: 24-29 | Draft len: 6

AR verify logits (top-5, sorted by AR)



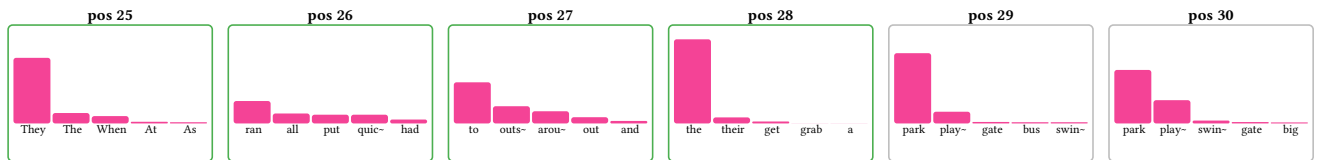
Diff draft logits (same tokens, AR order)



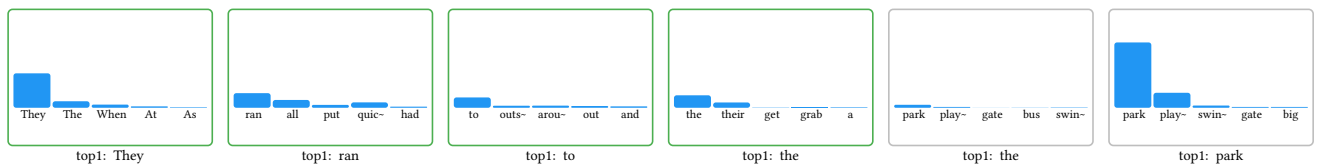
Sentence context Suddenly, he heard a loud noise.

Prompt: "One day, the teacher said" | Checkpoint: "step_0121566.npz" Target position: 29 | Block: 25-30 | Draft len: 6

AR verify logits (top-5, sorted by AR)



Diff draft logits (same tokens, AR order)



Sentence context They ran to the park and saw a big slide.